

## Easy (10 MCQs)

1.

**What does OpenFlow primarily enable?**

- a) Communication between control and data planes
- b) Wireless network communication
- c) Physical cabling
- d) DNS resolution

**Answer:** a

**Explanation:** OpenFlow is a protocol that allows a centralized controller to communicate with forwarding devices.

2.

**Which university is credited with originating OpenFlow?**

- a) MIT
- b) Stanford
- c) Harvard
- d) Berkeley

**Answer:** b

**Explanation:** OpenFlow was introduced by Stanford University.

3.

**In SDN, where is the control plane located?**

- a) Distributed in each switch
- b) Centralized in a controller
- c) Only in routers
- d) In the cloud only

**Answer:** b

**Explanation:** SDN centralizes the control plane in a controller for programmability.

4.

**What does the data plane do?**

- a) Makes forwarding decisions
- b) Runs network applications
- c) Manages routing protocols
- d) None of the above

**Answer:** a

**Explanation:** The data plane forwards packets according to rules set by the control plane.

5.

**OpenDaylight is:**

- a) A hardware switch
- b) An open-source SDN controller platform
- c) A routing protocol
- d) A firewall

**Answer:** b

**Explanation:** OpenDaylight is a modular open-source SDN controller.

6.

**Which of these is NOT a part of OpenDaylight architecture?**

- a) Southbound plugins
- b) Northbound APIs
- c) Physical routers only
- d) Controller core

**Answer:** c

**Explanation:** Physical routers are not architecture components but devices managed by the controller.

7.

**The control plane in traditional networks is:**

- a) Separate from the data plane

- b) Integrated into network devices
- c) Always centralized
- d) Part of the internet only

**Answer:** b

**Explanation:** Traditionally, the control and data planes reside together in network devices.

8.

**Which protocol is commonly used as a southbound interface in SDN?**

- a) HTTP
- b) OpenFlow
- c) FTP
- d) SMTP

**Answer:** b

**Explanation:** OpenFlow is the common protocol enabling controller-device communication.

9.

**The separation of control and data planes allows:**

- a) Faster innovation and easier management
- b) Increased cabling
- c) Lower security
- d) None of the above

**Answer:** a

**Explanation:** This separation enables programmability and centralized control.

10.

**OpenFlow rules are installed on:**

- a) Controllers
- b) Network switches
- c) End-user devices
- d) Firewalls

**Answer:** b

**Explanation:** Controllers install flow rules on OpenFlow-enabled switches.

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## Medium (30 MCQs)

11.

**Which version of OpenFlow introduced group tables for complex forwarding?**

- a) 1.0
- b) 1.1
- c) 1.3
- d) 2.0

**Answer:** c

**Explanation:** OpenFlow 1.3 added group tables and other enhancements.

12.

**What is the function of a flow table in OpenFlow?**

- a) Store forwarding rules
- b) Store routing protocols
- c) Store IP addresses
- d) None of the above

**Answer:** a

**Explanation:** Flow tables hold packet processing rules.

13.

**In OpenDaylight, southbound plugins:**

- a) Communicate with network devices

- b) Expose APIs to applications
- c) Run on end-user devices
- d) None of the above

**Answer:** a

**Explanation:** Southbound plugins connect the controller to devices.

14.

**Which OpenFlow message is used by switches to report new packets to the controller?**

- a) Packet-In
- b) Flow-Mod
- c) Hello
- d) Echo Request

**Answer:** a

**Explanation:** Packet-In messages notify the controller about packets requiring instructions.

15.

**OpenDaylight's northbound API allows:**

- a) Applications to interact with the network controller
- b) Controllers to connect to switches
- c) Routers to forward packets
- d) None of the above

**Answer:** a

**Explanation:** Northbound APIs expose controller functions to apps.

16.

**Control plane separation enables:**

- a) Centralized decision-making
- b) Distributed routing only
- c) Manual configuration
- d) None of the above

**Answer:** a

**Explanation:** Centralized control improves network programmability.

17.

**Which OpenFlow message modifies flow entries in a switch?**

- a) Flow-Mod
- b) Packet-In
- c) Packet-Out
- d) Hello

**Answer:** a

**Explanation:** Flow-Mod messages install or delete flow entries.

18.

**OpenFlow supports which types of match fields?**

- a) Ethernet, IP, TCP/UDP headers
- b) Only Ethernet addresses
- c) Only IP addresses
- d) None of the above

**Answer:** a

**Explanation:** OpenFlow matches on various packet header fields.

19.

**What is the purpose of the OpenDaylight controller core?**

- a) Manage device communications and network topology
- b) Physically forward packets
- c) Provide hardware firewall
- d) None of the above

**Answer:** a

**Explanation:** The controller core manages devices and network state.

20.

**Which of these is a key benefit of SDN with OpenFlow?**

- a) Centralized management of network traffic
- b) Fixed hardware configurations only
- c) Manual CLI configuration for each device
- d) None of the above

**Answer:** a

**Explanation:** SDN centralizes traffic management via software.

21.

**Which statement about OpenFlow switch states is true?**

- a) The switch waits for instructions when a flow is unknown
- b) The switch ignores unknown packets
- c) The switch always forwards packets without rules
- d) None of the above

**Answer:** a

**Explanation:** The switch sends unknown packets to the controller for decision.

22.

**In OpenFlow, a flow entry contains:**

- a) Match fields, actions, counters
- b) IP addresses only
- c) Routing tables only
- d) None of the above

**Answer:** a

**Explanation:** Flow entries specify matching criteria, actions to take, and statistics.

23.

**What is OpenDaylight's role in an SDN deployment?**

- a) Central controller and management platform
- b) Physical switch replacement
- c) Firewall device only
- d) None of the above

**Answer:** a

**Explanation:** OpenDaylight acts as the controller managing the network.

24.

**Which of the following is a common use case for OpenFlow?**

- a) Traffic engineering and load balancing
- b) Email delivery
- c) Web hosting only
- d) None of the above

**Answer:** a

**Explanation:** OpenFlow allows dynamic traffic management.

25.

**OpenFlow-enabled switches:**

- a) Forward packets based on flow rules from the controller
- b) Perform routing protocols independently
- c) Only forward packets via MAC address tables
- d) None of the above

**Answer:** a

**Explanation:** The controller programs switches with forwarding rules.

26.

**What is a southbound interface in SDN?**

- a) Protocol used by controller to communicate with devices
- b) Interface between applications and controller
- c) User interface to network



d) None of the above

**Answer:** a

**Explanation:** Southbound interfaces connect controller and network devices.

27.

**Which protocol besides OpenFlow can be used as a southbound protocol?**

a) NETCONF

b) HTTP

c) FTP

d) SMTP

**Answer:** a

**Explanation:** NETCONF is another southbound protocol for device configuration.

28.

**Which part of the network does OpenDaylight control?**

a) Data plane devices via southbound protocols

b) End-user devices only

c) Only routers

d) None of the above

**Answer:** a

**Explanation:** OpenDaylight controls forwarding devices like switches.

29.

**The OpenFlow protocol operates at which layer of the network?**

a) Between control and data planes

b) Application layer only

c) Physical layer only

d) None of the above

**Answer:** a

**Explanation:** OpenFlow bridges the control and forwarding layers.

30.

**OpenFlow switches can:**

- a) Buffer packets temporarily while waiting for controller decisions
- b) Always forward packets without controller input
- c) Operate without flow tables
- d) None of the above

**Answer: a**

**Explanation:** Switches buffer unknown packets for controller instructions.

31.

**Which OpenDaylight component is responsible for topology discovery?**

- a) Controller core
- b) Southbound plugin
- c) Northbound API
- d) Application layer

**Answer: a**

**Explanation:** The controller core manages topology information.

32.

**In OpenFlow, what happens when a packet matches no flow entry?**

- a) The packet is sent to the controller via Packet-In message
- b) The packet is dropped silently
- c) The packet is broadcast to the network
- d) None of the above

**Answer: a**

**Explanation:** Unknown packets are sent to the controller for instructions.

33.

**Which of the following is NOT a function of OpenDaylight?**

- a) Network orchestration
- b) Packet forwarding
- c) Device management
- d) API exposure for applications

**Answer:** b

**Explanation:** Packet forwarding is done by the data plane devices, not ODL.

34.

**What is the purpose of northbound APIs in OpenDaylight?**

- a) Allow applications to program the network
- b) Control switches directly
- c) Handle physical connections
- d) None of the above

**Answer:** a

**Explanation:** Northbound APIs expose services to applications.

35.

**Which of the following is a key feature of OpenFlow?**

- a) Flow-based forwarding rules
- b) Static routing only
- c) No programmability
- d) None of the above

**Answer:** a

**Explanation:** OpenFlow enables dynamic flow rules.

36.

**Which type of device is typically OpenFlow-enabled?**

- a) Switches
- b) Printers
- c) Laptops

d) None of the above

**Answer:** a

**Explanation:** Switches are typically OpenFlow-enabled for SDN.

37.

**OpenDaylight supports which programming languages for applications?**

a) Java, Python, and others

b) Only C++

c) Only Assembly

d) None of the above

**Answer:** a

**Explanation:** ODL supports multiple languages, Java being primary.

38.

**OpenFlow protocol defines messages exchanged between:**

a) Controller and switch

b) Two switches only

c) User and server

d) None of the above

**Answer:** a

**Explanation:** OpenFlow messages are between controller and switch.

39.

**OpenFlow supports:**

a) Both IPv4 and IPv6 packets

b) Only IPv4 packets

c) Only TCP packets

d) None of the above

**Answer:** a

**Explanation:** OpenFlow supports IPv4 and IPv6 matching.

40.

**In OpenDaylight, applications can be:**

- a) Network services like firewall, load balancer
- b) Only OS-level tools
- c) Hardware only
- d) None of the above

**Answer:** a

**Explanation:** Applications implement network functions on top of ODL.

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## Hard (10 MCQs)

41.

**What is the function of a “Group Table” in OpenFlow 1.3?**

- a) Supports multipath forwarding and fast failover
- b) Stores IP addresses
- c) Manages VLAN tags only
- d) None of the above

**Answer:** a

**Explanation:** Group tables enable advanced forwarding techniques like load balancing.

42.

**How does OpenDaylight handle device abstraction?**

- a) Via southbound plugins that translate device-specific protocols to a common model
- b) Through northbound APIs only
- c) Using static configurations

d) None of the above

**Answer:** a

**Explanation:** Southbound plugins abstract device specifics.

43.

**Which OpenFlow message allows the controller to send packets out of the switch without installing flow entries?**

a) Packet-Out

b) Flow-Mod

c) Packet-In

d) Hello

**Answer:** a

**Explanation:** Packet-Out messages instruct the switch to send a packet immediately.

44.

**In OpenFlow, the cookie field in a flow entry is used for:**

a) Identifying flows for management or statistics

b) Storing packet payload

c) Encrypting traffic

d) None of the above

**Answer:** a

**Explanation:** Cookies help track flows for monitoring or deletion.

45.

**OpenDaylight uses which data store technology internally?**

a) MD-SAL (Model-Driven Service Abstraction Layer)

b) SQL database only

c) Flat files only

d) None of the above

**Answer:** a

**Explanation:** MD-SAL manages network state in ODL.

46.

**Which of these is NOT true about OpenFlow flow entries?**

- a) They can have timeouts for automatic removal
- b) They can match on packet payload content
- c) They specify actions for matched packets
- d) They keep counters for statistics

**Answer:** b

**Explanation:** OpenFlow matches header fields, not packet payload.

47.

**In OpenDaylight, how are network events propagated to applications?**

- a) Via event listeners on the MD-SAL data store
- b) Through manual polling only
- c) Via email notifications
- d) None of the above

**Answer:** a

**Explanation:** Applications listen to changes in the data store.

48.

**Which of the following best describes the OpenFlow pipeline?**

- a) Multiple flow tables evaluated in sequence
- b) A single static routing table
- c) Only hardware-based forwarding
- d) None of the above

**Answer:** a

**Explanation:** OpenFlow supports multiple tables in a pipeline.

49.

**How does OpenDaylight achieve scalability?**

- a) Modular architecture and clustering support
- b) Single monolithic design only
- c) Only supports small networks
- d) None of the above

**Answer:** a

**Explanation:** ODL supports clustering and modular components.

50.

**Which southbound protocol besides OpenFlow is natively supported by OpenDaylight for device configuration?**

- a) NETCONF
- b) SMTP
- c) FTP
- d) HTTP

**Answer:** a

**Explanation:** NETCONF is used for device management.